In-vitro Biology

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Cell engineering Workflow

Cell engineering

Different strategies for different applications

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Membrane receptor cell lines
(Low, Medium and High expressing clones)

KI/KD/KO cell lines
(Flp-In, siRNA / shRNA & CRISPR/Cas9)

Inducible cell line
(Tet / Cumate inducible system)

Reporter cell lines
(Fluorescent protein/ Luciferase)

Production cell lines
(Freedom CHO-S, CHO DG44 & Lentiviral vector)

Vector systems

(pcDNA series, pLVX and pGIPZ / pTRIPZ lentiviral vector series, pCHO1.0, pOptivec, pGL and in-house modified vector systems)

Transfection methods

Standard (Lipid / nanoparticle) Electroporation, Transposon and Lentiviral transduction

Screening

(Flow cytometry, Western blot, PA-HPLC, ELISA Octet and ClonePix)

QC

Thaw test, Mycoplasma, Clonal stability, Receptor density estimation,
SE-HPLC and LC-MS analysis

Case study: Knock-In (KI) cell line generation

Site specific knock-in of a target gene using CRISPR/Cas9 technology

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gRNA (3 – 5 gRNA per target gene) and ssDonor template designing for KI

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Transfection of host cell line using gRNA, Cas9 protein (RNP complex) and ssDonor DNA for KI

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Detection of gene editing using western blot / flowcytometry analysis

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Single cell cloning followed by WB / Flow cytometry analysis for KO / KI

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KO / KI validation using Sanger sequencing

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QC (Thaw test, Mycoplasma)

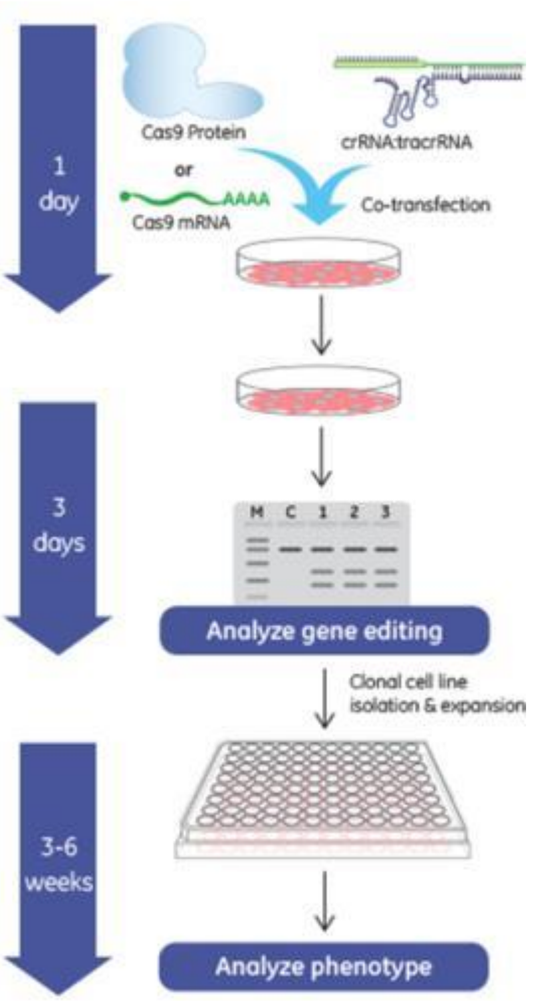
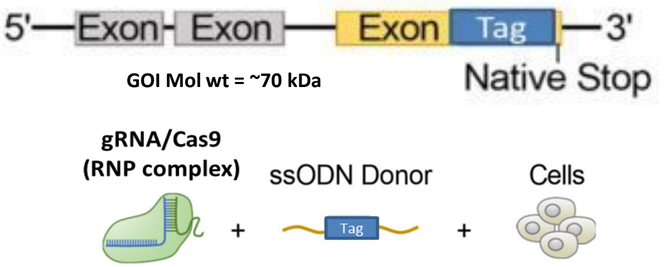
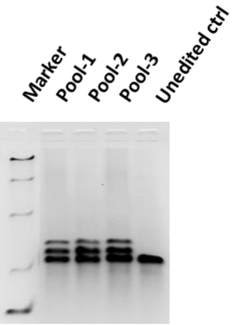


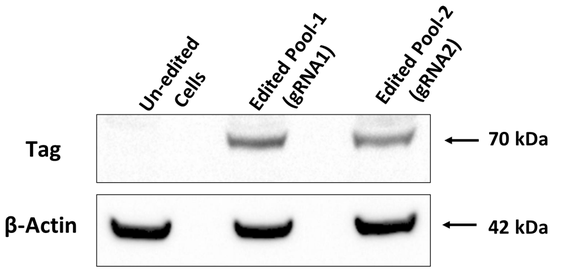
Image: Dharmacon

Case study: Knock-in of 11 aa HiBiT tag in the last exon of an endogenous gene of interest using CRISPR/Cas9

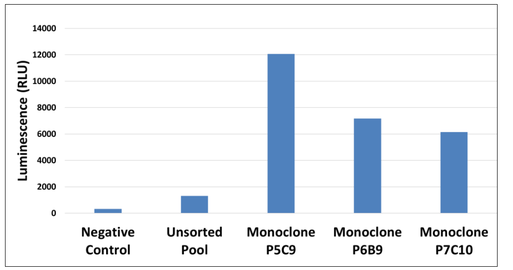




Gene editing confirmation by PCR



Western blot analysis to check the expression of tagged protein in the CRISPR/Cas9 edited cells using anti-tag antibody



Luminescence signals in gene edited HiBiT tag KI monoclones

Case study: Knock-down (KD) cell line generation

Inducible shRNA

3 – 5 siRNA /shRNA designing per target for KD

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Transfection of host cell line using siRNAs or shRNAs (lentiviral system)

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Detection of target gene KD by western blot / qRT-PCR / flow cytometry methods

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Single cell cloning (for shRNA stable cell lines generation)

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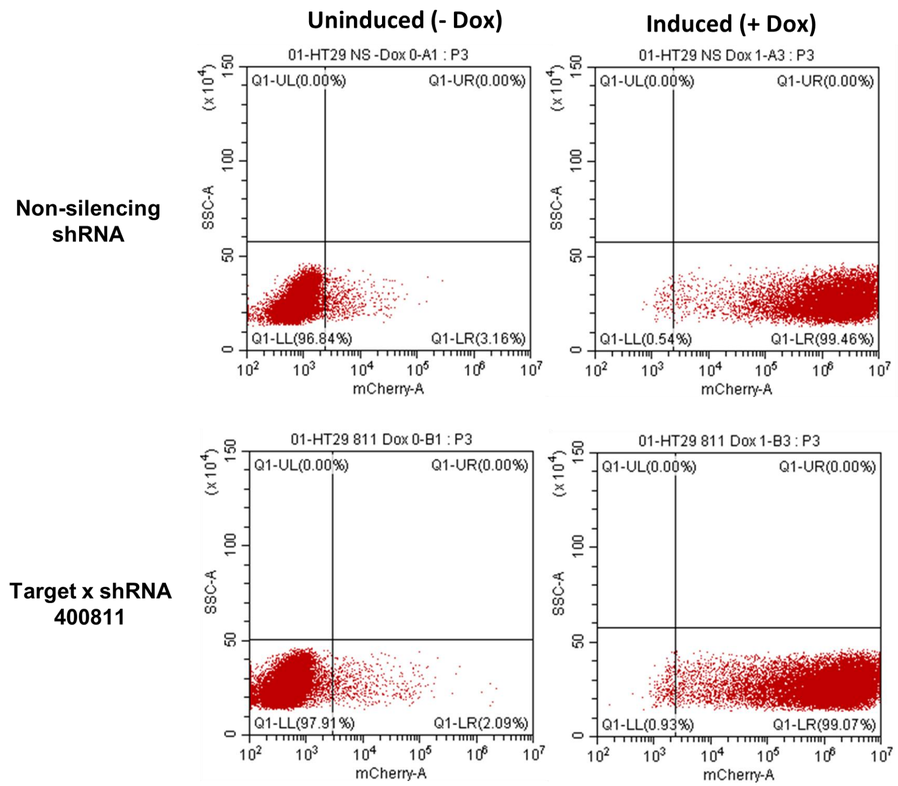
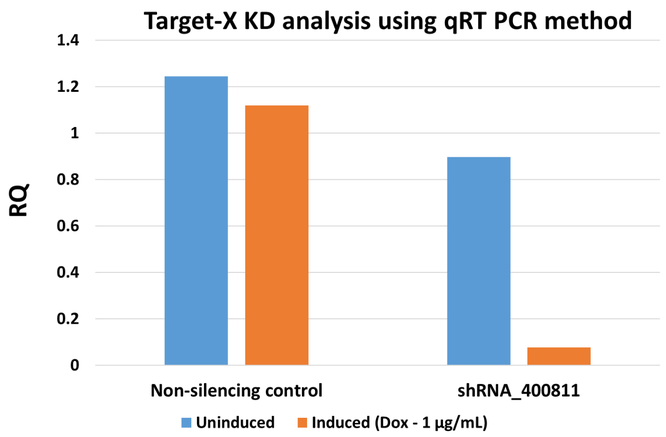
KD validation by WB /qRT-PCR / flow cytometry analysis

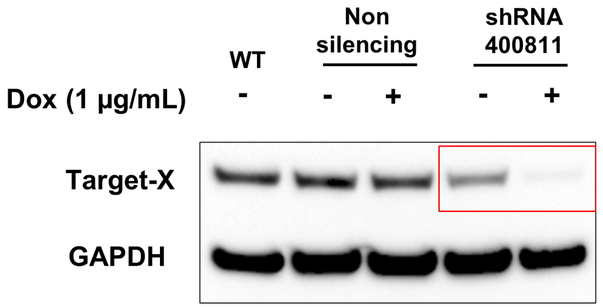
↓

In vitro / functional assay)

Case study: Doxycycline inducible shRNA mediated downregulation of target-X in a cancer cell line







Case study: Generation of LgBiT and HiBiT tagged Target-Y expressing reporter cell line for HiBiT assay

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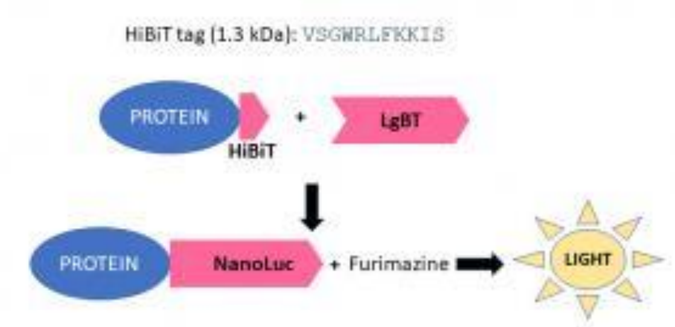
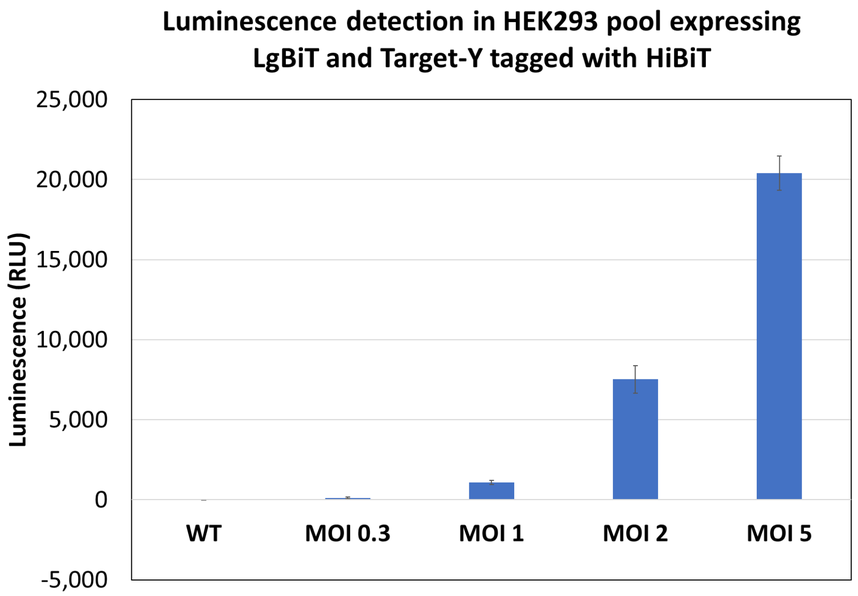
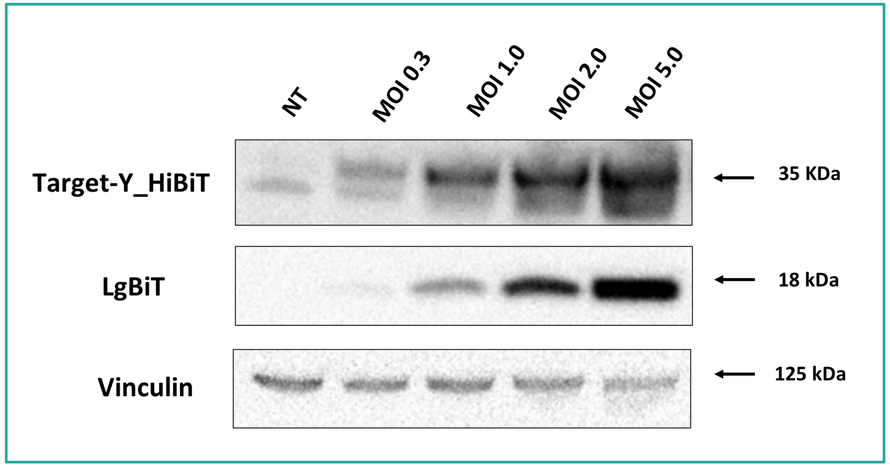
HiBiT and LgBiT system



HiBiT is a 11 amino acid small peptide that binds with high affinity to another larger subunit called LgBiT. The bound complex has luciferase activity and will release luminescent signal in the presence of added furimazine substrate





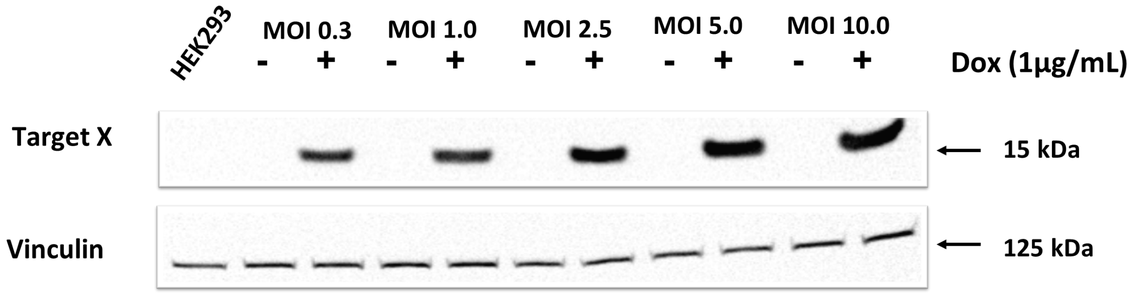
Case study: Inducible cell lines

Tet / cumate inducible system

APSL has extensive experience working with various vector / cell line systems for generating inducible cell lines:

- **T-REx-293, T-REx-HeLa, T-REx-CHO and T-REx-Jurkat inducible cell line system (Thermo):** T-REx is a Tet-ON system limited to host cell lines which stably express the tetracycline repressor protein
- **pLVX-TetOne lentiviral system (Takara):** This is a Tet-ON system contains all the necessary components in a single vector and can be used in any host cell line
- **pTRIPz lentiviral system (Horizon):** Tet-ON lentiviral single vector system
- **SparQ All-in-One lentivector system (SBI System Biosciences) :** Cumate inducible system

Case study: Generation of Target X expressing inducible cell line



An increase in Target X expression was seen in HEK293 cell line transduced at different MOI after doxycycline induction